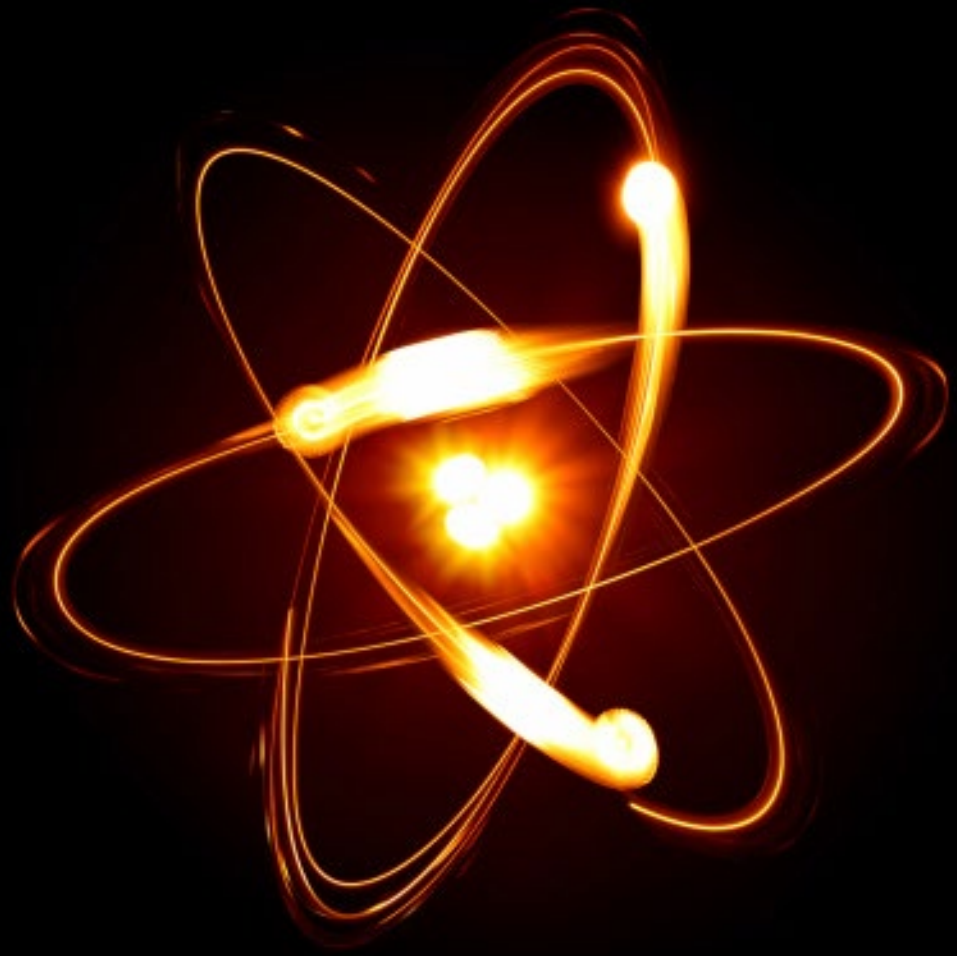
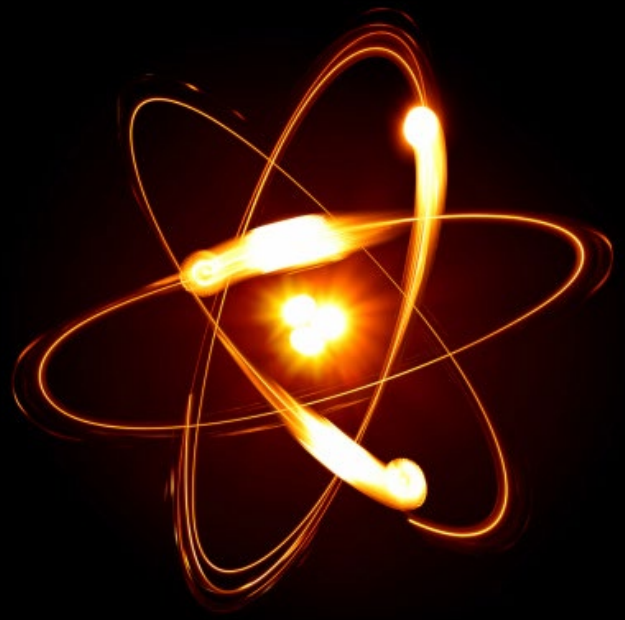
Chapter 20.

Spin 1/2



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Quantum Physics



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Spin 1/2

- Commutation relations
- Basis of spin 1/2
- Matrix representation
- Eigenvalues and eigenvectors
- Component along general direction $\vec{u}$
- Postulates of Pauli Theory

Commutation relations.

Let $\vec{S}$ denote the electron's spin, such that S^2 has eigenvalues $\frac{1}{2}\left(\frac{1}{2}+1\right)\hbar^2 = \frac{3}{4}\hbar^2$, and correspondingly, S_z has eigenvalues $\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$

Since $\vec{S}$ is an angular momentum we also have:

$$[S_x, S_y] = i\hbar S_z$$

$$[S_y, S_z] = i\hbar S_x$$

$$[S_z, S_x] = i\hbar S_y$$

$$[S^2, S_x] = [S^2, S_y] = [S^2, S_z] = 0$$

Basis of spin 1/2.

A basis of the spin state space, $\mathcal{E}_s$, which describes the electron's spin state is given by the eigenvectors of S^2 and S_z : (it is postulated that there is no degeneracy so dim $\mathcal{E}_s = 2$)

$$\left\{\left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle, \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle\right\} = \{|+\rangle, |-\rangle\}$$

Some properties:

- The basis is orthonormal: $\langle+|-\rangle = \langle-|+\rangle = 0$ $\langle+|+\rangle = \langle-|-\rangle = 0$
- Closure relation: $|+\rangle\langle+| + |-\rangle\langle-| = \mathbb{I}$
- $S_+|+\rangle = 0$
- $S_+|-\rangle = \hbar|+\rangle$
- $S_-|-\rangle = 0$
- $S_-|+\rangle = \hbar|-\rangle$

Matrix representation

Let us compute the matrices that represent, in the previous basis, the operators S^2, S_z, S_y and S_x

We have the basis vectors as: $\{|+\rangle, |-\rangle\}$

Then, the action of S^2 on the basis vectors is (they are eigenvectors)

$$S^2|+\rangle = \frac{3}{4}\hbar^2|+\rangle \qquad S^2|-\rangle = \frac{3}{4}\hbar^2|-\rangle$$

Therefore the matrix that represents S^2 is:

$$S^2 = \frac{3\hbar^2}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Matrix representation

Since the basis vectors are also eigenvectors of S_z , the matrix representing the operator is:

$$S_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

To compute the matrix that represents S_x we calculate

$$S_x|+\rangle = \alpha_+|+\rangle + \beta_+|-\rangle \qquad S_x|-\rangle = \alpha_-|+\rangle + \beta_-|-\rangle$$

and the matrix will be:

$$S_x = \begin{pmatrix} \langle +|S_x|+\rangle & \langle +|S_x|-\rangle \\ \langle -|S_x|+\rangle & \langle -|S_x|-\rangle \end{pmatrix} \quad \Rightarrow \quad S_x = \begin{pmatrix} \alpha_+ & \alpha_- \\ \beta_+ & \beta_- \end{pmatrix}$$

Matrix representation

From the definition of S_+ and S_-

$$S_x = \frac{1}{2}(S_+ + S_-) \qquad S_x|+\rangle = \frac{1}{2}(S_+ + S_-)|+\rangle = \frac{\hbar}{2}|-\rangle \qquad S_x|-\rangle = \frac{\hbar}{2}|+\rangle$$

therefore

$$S_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Similarly, using that $S_y = \frac{1}{2i}(S_+ - S_-)$

$$S_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Eigenvalues and eigenvectors

Let us find the eigenvalues and eigenvectors of S_x and S_y.

A state, $|\phi\rangle_x$, belonging to the state space, $\mathcal{E}$, is an eigenstate of S_x if

$$S_x|\phi\rangle_x = \lambda|\phi\rangle_x$$

Writing this explicitly in the basis $\{|+\rangle, |-\rangle\}$ we have

$$\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \lambda\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$$

that is:

$$\begin{pmatrix}-\lambda & \hbar/2\\ \hbar/2 & -\lambda\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix} \qquad (1)$$

Eigenvalues and eigenvectors

Equation (1) in the previous slide has non-trivial solutions if

$$\begin{vmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{vmatrix} = 0 \qquad \Longrightarrow \quad \lambda = \pm\,\hbar/2$$

For $\lambda = \hbar/2$ the normalized eigenstate is (taking into account (1) and the normalization condition $|\alpha|^2 + |\beta|^2 = 1$):

$$|+\rangle_x = \frac{1}{\sqrt{2}}[|+\rangle + |-\rangle]$$

while for $\lambda = -\,\hbar/2$

$$|-\rangle_x = \frac{1}{\sqrt{2}}[|+\rangle - |-\rangle]$$

Eigenvalues and eigenvectors

For S_y , following a similar procedure, one finds:

- Eigenvalues: $\pm\frac{\hbar}{2}$
- Eigenstates:

$$|+\rangle_y = \frac{1}{\sqrt{2}}\left[|+\rangle + i|-\rangle\right]$$

$$|-\rangle_y = \frac{1}{\sqrt{2}}\left[|+\rangle - i|-\rangle\right]$$

Component along general direction $\vec{u}$

If we consider the spin component along the direction of a unit vector, $\vec{u}$, pointing according to the angles (θ, ϕ) then

$$\vec{u} = \sin\theta\cos\phi\,\vec{\imath} + \sin\theta\sin\phi\,\vec{\jmath} + \cos\theta\,\vec{k}$$

On the other hand, $\vec{S} = S_x\vec{\imath} + S_y\vec{\jmath} + S_z\vec{k}$, so

$$S_u = \vec{S}\cdot\vec{u} = S_x\sin\theta\cos\phi + S_y\sin\theta\sin\phi + S_z\cos\theta$$

(in the basis $\{|+\rangle, |-\rangle\}$.)

$$S_u = \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\phi} \\ \sin\theta\, e^{i\phi} & -\cos\theta \end{pmatrix}$$

Component along general direction $\vec{u}$

For S_u, following a similar procedure, one finds:

- Eigenvalues: $\pm\frac{\hbar}{2}$
- Eigenstates:

$$|+\rangle_u = \cos\frac{\theta}{2}e^{-i\phi/2}|+\rangle + \sin\frac{\theta}{2}e^{i\phi/2}|-\rangle$$

$$|-\rangle_u = -\sin\frac{\theta}{2}e^{-i\phi/2}|+\rangle + \cos\frac{\theta}{2}e^{i\phi/2}|-\rangle$$

The operators S_x, S_y, S_z and S_u have the same eigenvalues $\hbar/2$ and $-\hbar/2$.

This is so because **all directions of space have the same properties.**

Postulates of Pauli Theory

The spin of an electron emerges naturally in a relativistic theory (Dirac equation). However, in a non-relativistic theory like ours, it must be postulated.

1. Each particle has an intrinsic angular momentum (or spin). This is described by a spin operator, $\vec{S}$, whose three components (S_x, S_y, S_z) satisfy the corresponding commutation relations.

2. Spin operators act only in the spin state space $\mathcal{E}_s$, where S^2 and S_z form a CSCO. The space $\mathcal{E}_s$ is spanned by the set of eigenstates $|s,m\rangle$ common to S^2 and S_z :

$$S^2|s,m\rangle = s(s+1)\hbar^2|s,m\rangle \qquad S_z|s,m\rangle = m\hbar\,|s,m\rangle$$

   and whose dimension is (2s + 1).

3. The state space of a particle with spin is the tensor product space $\mathcal{E} = \mathcal{E}_r \otimes \mathcal{E}_s$. Therefore, all spin observables commute with all orbital observables.

4. Contrary to the orbital magnetic dipole moment, which is $\vec{\mu}_L = \frac{\mu_B}{\hbar}\vec{L}$, with $\mu_B = \frac{q\hbar}{2m_e}$, the spin magnetic dipole moment is $\vec{\mu}_s = 2\frac{\mu_B}{\hbar}\vec{S}$. The value for the gyromagnetic spin factor is also derived in a relativistic theory.